

Linear Regression — 60-Minute Session Walkthrough

Oct 6, 2026 · @Scott

Session at a glance

The hour is a recorded video: three talk-to-camera blocks wrapped around one screen-recorded marimo demo, so viewers see every formula on the slides move on screen. The deck (M10_01, CAI5735) covers the regression line, slope and intercept, residuals, least squares, and model evaluation; the demo makes each of those visible with a BMI → systolic BP dataset that matches the slides' running example. With no live audience, questions become **pause prompts**: pose the question, say "pause the video and try it," wait two seconds, then answer it yourself.

Learning objectives (from slide 2): by the end, students can read $\hat{y} = a + bx$ in clinical units, interpret slope and intercept, compute and interpret residuals, and explain why least squares picks one unique line.

Materials

- Slide deck: M10_01_CAI5735_su26_Aggarwal_v1 (12 slides)
- Notebook: `linear_regression_demo.py` (marimo), run with `marimo edit linear_regression_demo.py`, or `marimo run` for a locked presentation view
- Required reading: [Inferential Thinking, Ch. 15 — Prediction](#)

Timing plan (60 min)

Clock	Min	Segment	Slides	Format
0:00	3	Open: objectives and a prediction hook	1-2	To camera + pause prompt
0:03	10	The regression line, slope and intercept	3-4	To camera
0:13	12	Demo part 1: drag the line, watch residuals and SSR	5-6 (preview)	Screen recording
0:25	10	Residuals and least squares	5-6	To camera

Clock	Min	Segment	Slides	Format
0:35	10	Evaluating the model: R^2 , RMSE, residual plots	7	To camera + screen recording
0:45	7	Regression in clinical practice	8	To camera
0:52	5	Python in practice: the three-step recipe	9	Screen recording (code)
0:57	3	Agenda, assignment, check questions	10–12	To camera

Record each row as its own take. Short takes are easier to redo, and the clock column doubles as chapter markers if your video platform supports them.

The demo is deliberately placed before the residuals slide: you fit a line by hand on screen first (viewers can pause and try their own in the notebook), so the least squares slide answers a question the video has already raised. If time runs short, compress slide 8 to two examples (HbA1c and gestational age) and keep the demo intact.

Recording script

The full spoken script, in the order you'll record it. Each heading is one slide or one notebook section, with its start time from the timing plan. Under each heading, **On screen** says what viewers see, the quoted text is what you say, and [brackets] are cues. Numbers in the notebook sections match the default settings (noise 10, first sample).

0:00 · Slide 1 — Title and hook (1.5 min)

On screen: title slide. Look at the lens for the hook question.

Hi everyone, and welcome to Module 10. I'm [name], and today we're covering linear regression, which is probably the most widely used tool in all of health research.

Let's start with a question. Imagine a new patient walks into clinic. Their BMI is 28. Before anyone puts a cuff on their arm, what's your best guess for their systolic blood pressure? Pause the video and write down a number.

[Pause 2 seconds.]

Hold on to that guess. We'll come back to it at the end.

Whatever number you wrote down, you probably used some intuition: heavier patients tend to have higher blood pressure. Regression turns that intuition into something precise. It draws one straight line through the data that tells us, for any BMI, what blood pressure we'd predict, and just as importantly, how far off that prediction is likely to be.

Along the way, I'll switch over to an interactive Python notebook built in marimo. It's posted alongside this video, so if you'd like, open it now and follow along. Whenever I say "pause and try it," that's your cue. The fastest way to understand regression is to drag a line around yourself and watch what happens.

0:02 · Slide 2 — Objectives (1.5 min)

On screen: the four objectives.

Here's where we're headed. By the end of this video, you should be able to do four things.

First, read a regression equation, $\hat{y} = a + bx$, and say what it means for patients, in real clinical units. Second, interpret the slope and the intercept, and know when the intercept is meaningful and when it isn't. Third, work with residuals: how far off each prediction is, and what a big one is telling you. And fourth, explain the least squares method: why, out of every line you could draw through a cloud of points, one specific line counts as the best.

After that, we'll look at how to judge whether that best line is any good, and at how regression shows up in clinical practice. Let's get started.

0:03 · Slide 3 — What is linear regression? (5 min)

On screen: the equation $\hat{y} = a + b \cdot x$ and the BMI vs. systolic BP scatter.

Linear regression fits a straight line through a scatter of data points. On one axis we have a predictor, x . Here that's BMI. On the other we have a response, y , which needs to be a continuous number. Here that's systolic blood pressure.

The line has an equation: $\hat{y}$ equals a plus b times x . Let's take it one piece at a time. x is the predictor value, a patient's BMI. b is the slope: how much our prediction changes for each one-unit increase in x . a is the intercept: the predicted value when x equals zero. And $\hat{y}$ is the prediction itself.

I want to stop on that little hat, because the whole lesson depends on it. y is what we actually measured for a patient. $\hat{y}$ is what the line predicts. They're almost never the same, and the gap between them is going to become one of our main characters in a few minutes.

[Pause prompt.] Here's a question to pause on. Suppose BMI and blood pressure were completely unrelated. What would the best prediction line look like?

[Pause 2 seconds.]

It would be flat. If BMI tells you nothing, the best you can do is predict the average blood pressure for everyone. A line with a slope is a claim that x carries some information about y .

0:08 · Slide 4 — Interpreting slope and intercept (5 min, including the transition)

On screen: $\hat{y} = 68.5 + 2.4 \cdot \text{BMI}$, with the slope and intercept panels.

Here's a fitted line for our example: predicted systolic blood pressure equals 68.5 plus 2.4 times BMI.

Start with the slope, 2.4. The units matter here: it's 2.4 millimeters of mercury per one kilogram-per-meter-squared of BMI. In plain language, patients whose BMI is one unit higher are predicted to have a systolic pressure about 2.4 points higher.

Notice how I phrased that. A common mistake is to read the slope as "if a patient's BMI goes up by one, their blood pressure goes up by 2.4." The line doesn't tell us that. It compares different patients; it doesn't follow one patient over time, and it doesn't prove that BMI causes anything.

Now the intercept, 68.5. Technically, it's the predicted blood pressure at a BMI of zero. But nobody has a BMI of zero. The intercept is a mathematical anchor that positions the line. It isn't a clinical estimate. Predicting that far outside your data is called extrapolation, and we'll see later why it's risky.

Let's use the line. A patient with a BMI of 28: 68.5 plus 2.4 times 28 gives 135.7 millimeters of mercury.

[Pause prompt.] Try this one yourself. Two patients differ in BMI by 5 units. How far apart are their predicted blood pressures?

[Pause 2 seconds.]

Twelve points. Five times the slope of 2.4. The intercept drops out completely.

So far, this line has just appeared on the slide with its numbers already filled in. But where did those numbers come from? Out of all the lines you could draw through a cloud of patients, why that one? Instead of telling you, I'd like to show you. I'm switching over to the notebook now.

[Cut to the notebook, section 0.]

0:13 · Notebook §0 — Meet the data (2 min)

On screen: `marimo run` view, scrolled to "0 · Meet the data": the noise slider, the resample button, and the first five rows.

This is our notebook. What you're looking at is a simulated clinic: 200 patients, each with a BMI and a systolic blood pressure. I built it from the same equation we just saw, plus some random noise, so we know the true answer and can check whether our methods find it.

Here are the first few patients. Patient one has a BMI of 24.1 and a pressure of 131. Patient two, 17.1 and 109.

Up here are two controls I'll use later. The noise slider sets how scattered the patients are around the true line. The button draws a brand-new clinic sample. Leave both alone for now.

One thing to notice: BMI in this clinic runs from about 17 to 42. Keep that range in mind. It's going to matter at the end.

[Scroll to section 1.]

0:15 · Notebook §1 — Your line vs. the computer's (7 min)

On screen: the intercept and slope sliders, the scatter plot with the orange line and red residual segments, and the score tiles. Start from the defaults ($a = 110$, $b = 0.5$).

Here's the game. These two sliders control a line: one for the intercept, one for the slope. The orange line is mine, and right now it's pretty bad. It's too flat.

Look at the red segments. Each one connects a patient to my line, straight up or straight down. That's the gap between what we measured and what my line predicts. Notice they're vertical, not the shortest distance to the line. We're predicting blood pressure, so we measure our errors in blood pressure.

To score the line, the notebook squares each of those gaps and adds them up. That total is called the sum of squared residuals, or SSR. Lower is better. My SSR right now is about 55,700, and the tile says that's more than three times the minimum possible.

Also look at this tile: the mean of my residuals is plus 11.8. On average, my line sits about 12 points below the patients. That's a clue that I should raise it, and tilt it up.

[Adjust the slope up toward 2 and the intercept down toward 80, narrating as the red segments shrink and the SSR falls.]

Watch the red lines shrink as I tilt it up. The SSR is falling. As I steepen the slope, I need to bring the intercept down to keep the line in the middle of the cloud. The two numbers trade off against each other.

[Settle on a line close to the best one; read the SSR and the ratio aloud.]

I'm at about [read the ratio] times the minimum. I think that's as close as I can get by eye.

[Pause prompt.] Now it's your turn. If you have the notebook open, pause the video and see if you can get within 5% of the minimum, before you reveal the computer's answer.

[Pause 2 seconds.]

[Tick "Reveal the least squares line".]

And here's the computer's line, the dashed navy one: $\hat{y} = 76.3 + 2.11 \times \text{BMI}$. Its SSR is about 16,800. Notice the black X: that's the point where average BMI meets average blood pressure, and the computer's line goes right through it. That's not a coincidence, and we'll come back to it.

[Scroll to section 2.]

0:22 · Notebook §2 — Why least squares picks one line (3 min, including the transition)

On screen: the SSR-vs-slope parabola, with the navy minimum and the orange dot for your slope.

So why couldn't I beat it? This plot shows why. I've pinned the line so it always passes through that point of averages, and then tried every slope from minus 2 to plus 6. For each one, I computed the SSR.

The result is a bowl. A parabola. And a bowl has exactly one bottom. That bottom, at a slope of 2.11, is the least squares line.

[Drag the slope slider back and forth.]

When I move my slope, the orange dot climbs up the side of the bowl. Any other slope, steeper or flatter, gives a bigger total error. That's what makes the least squares line "the" best line, not just a good one.

Those red gaps we've been looking at have a name: residuals. Let's go back to the slides, define them properly, and see why squaring them is the key to all of this.

[Cut back to slide 5.]

0:25 · Slide 5 — Residuals and prediction errors (5 min)

On screen: $e = y - \hat{y}$ and the residual diagram.

A residual is the difference between what we observed and what the line predicted: e equals y minus $\hat{y}$. Those are exactly the red segments from the notebook.

The sign tells you which way the line missed. A positive residual means the patient sits above the line: their real blood pressure is higher than predicted, so the model underpredicted. A negative residual means the patient sits below the line, and the model overpredicted.

Here's a property that's always true for a least squares line: the residuals add up to zero. The overpredictions and underpredictions balance out exactly. Remember the tile in the notebook that said the mean of my residuals was plus 11.8? For the computer's line, that number is zero. A line that's systematically too high or too low can always be improved by sliding it.

Residuals also have a clinical use. A patient with a very large residual is someone the model explains poorly. That's worth a closer look.

[Pause prompt.] Pause and think about this one. A patient's residual is plus 22 millimeters of mercury. What would you tell the clinician?

[Pause 2 seconds.]

Their blood pressure is much higher than their BMI alone would predict. Something else is going on, whether that's medication, kidney function, or something else, and BMI isn't capturing it.

0:30 · Slide 6 — The least squares method (3 min)

On screen: $SSR = \sum (y_i - \hat{y}_i)^2 = \sum e_i^2$, the four boxes, and the `linregress` line.

In the notebook, I couldn't beat the computer's line. Here's why nobody can. The least squares line is defined as the one line that makes the sum of squared residuals as small as possible. We saw that as the bottom of the bowl.

Why square the residuals instead of just adding them up? Three reasons. First, if we just added them, positive and negative errors would cancel, and a terrible line could score zero. Squaring makes every error count. Second, squaring punishes big misses much more than small ones: an error of 20 counts four times as much as an error of 10. And third, squaring gives us a smooth bowl, so the minimum can be found exactly with a formula.

Here's that formula. The slope equals r , the correlation between x and y , times the standard deviation of y divided by the standard deviation of x . The intercept is the average y minus the slope times the average x . That second formula is why the line always passes through the point of averages, the black X we saw.

There's a cost to squaring, though. Because big errors count so heavily, one extreme patient can pull the whole line toward them. We'll see that happen in a few minutes.

In Python, you don't compute any of this by hand. One function, `stats.linregress`, returns the slope, the intercept, the correlation r , and two more values, a p-value and a standard error, that we'll leave for another day. Let me show you that the formula and the function really do agree.

[Cut to notebook section 3.]

0:33 · Notebook §3 — Same line, two ways (2 min)

On screen: the formula and the comparison table.

On the left, I computed the line by hand: convert both variables to standard units, average their products to get r , then plug into the slope and intercept formulas. On the right is what `linregress` returns. The correlation is 0.729, the slope is 2.11, and the intercept is 76.3, and the two columns agree to four decimal places.

Now, notice something. I built this clinic from a true slope of 2.4, but we estimated 2.11. Why? Because 200 patients is a sample, and every sample is a little different.

[Click "Draw a new clinic sample" two or three times.]

Watch the slope as I draw new clinics. It wobbles around 2.4. Any line you fit from data is an estimate. How much those estimates wobble is a question for another lesson, but it's worth knowing that they do.

[Refresh the page to restore the first sample, then cut back to slide 7.]

0:35 · Slide 7 — Evaluating the regression model (4 min)

On screen: the three panels: R^2 , RMSE, and the residual plot.

We've found the best line. But "best" only means best among straight lines. It doesn't mean good. So after fitting a line, we always ask: how well does it actually predict?

Three tools help.

First, R-squared, the coefficient of determination. It's the share of the variation in y that the line accounts for, between zero and one. An R-squared of 0.52 means BMI accounts for 52% of the variation in systolic pressure across these patients; the other 48% comes from everything else. With a single predictor, R-squared is simply the correlation, r , squared.

Second, RMSE, the root mean squared error. Square the residuals, average them, take the square root. That puts the error back in the original units. An RMSE of 8 means the predictions are typically off by about 8 millimeters of mercury. R-squared tells you how much of the pattern you captured; RMSE tells you how big a miss to expect for a patient.

Third, the residual plot: residuals on the vertical axis, predicted values on the horizontal. If the line is a good summary, you'll see a shapeless band of points scattered evenly around zero. Any pattern in that plot means the line is missing something.

[Pause prompt.] Here's one to think about. A model has an R-squared of 0.25 and an RMSE of 14 millimeters of mercury. Would you use it to screen individual patients for hypertension?

[Pause 2 seconds.]

Probably not. A typical miss of 14 points is the difference between normal and stage 1 hypertension. The model might still describe a real trend across a population, but it isn't precise enough for individual decisions.

These numbers are much easier to understand when you can watch them change. Let's go back to the notebook.

[Cut to notebook section 4.]

0:39 · Notebook §4 — Evaluating the fit (4 min)

On screen: the scenario dropdown and outlier checkbox, the scatter and residual plots side by side, and the stat tiles. Start on "Linear (healthy)" with the outlier off.

Here's our clinic again with the least squares line on the left and its residual plot on the right. The residual plot is a shapeless band around zero, which is exactly what we want.

The tiles: R-squared is 0.53, so BMI accounts for about half the variation. RMSE is 9.2 millimeters of mercury. And look at this one: the square root of one minus r-squared, times the standard deviation of y , is also 9.2. That's a handy shortcut: if you know the correlation and the spread of y , you already know the RMSE. Finally, the mean residual is zero, as promised.

Now let's break things. Real data don't always follow a straight line.

[Switch the scenario to "Curved (nonlinear)".]

In this clinic, blood pressure rises faster at high BMI. The line still goes through the middle, and R-squared is about 0.40, which doesn't look terrible. But look at the residual plot. Positive on both ends, negative in the middle. A U shape. The line is systematically wrong in a predictable way. A common mistake is to see a decent R-squared and assume the line is right. Only the residual plot reveals this.

[Switch to "Fan (unequal spread)".]

Here's a different problem. The residuals fan out as predicted pressure rises. R-squared is about 0.52, almost the same as the healthy clinic. But our predictions are much less precise for high-BMI patients than for low-BMI patients, and a single RMSE hides that.

[Switch back to "Linear (healthy)" and tick "Add one extreme patient".]

Last one. I've added a single patient with a BMI of 44 and a pressure of 95, circled in red. One patient out of 201. Watch the line: the slope drops from 2.11 to 1.85. The gray dashed line shows where it was. Because errors are squared, this one patient's miss of nearly 80 points counts as much as 64 ordinary misses of 10 points, so the line bends toward them.

Should we delete this patient? Not automatically. First check whether it's a data entry error or a measurement problem. If it's real, report your results with and without it, so readers can see how much one point matters.

[Untick the outlier and scroll to section 5.]

0:43 · Notebook §5 — Make a prediction (2 min)

On screen: the BMI input and the prediction line below it.

Finally, let's use the line. I'll type in a BMI of 28, our patient from the start.

The prediction is 76.3 plus 2.11 times 28, which is 135.4 millimeters of mercury, with a rough range of 117 to 154. That range is plus or minus two RMSEs: a quick guide to where most patients with this BMI fall.

[Type 55.]

Now a BMI of 55. The line happily gives us a number. But a warning appears: our data only go up to about 42. We have no evidence about what happens out there. This is the same problem as the intercept: the line will always give you an answer, but that doesn't mean you should trust it.

[Cut back to slide 8.]

0:45 · Slide 8 — Regression in clinical practice (7 min)

On screen: the four clinical examples. This script assumes the slide has been corrected (see Slide fixes before recording); swap in your final figures where marked.

Let's step back from our simulated clinic and look at how regression shows up in real clinical work. As we go through these four examples, ask yourself one question: is this relationship strong enough to predict an individual patient, or does it only describe a trend across a population?

Start with the strongest. HbA1c and average blood glucose. In a large international study, a straight line from HbA1c to mean glucose had an R-squared of 0.84. That's strong enough that the American Diabetes Association turned it into a formula clinicians use every day to translate an A1c result into an estimated average glucose for patients. Here regression is a working clinical tool.

Next, gestational age and birth weight. [Insert your verified R^2 .] This relationship underlies the growth charts obstetricians use to flag babies who are small or large for their gestational age. One caution: across the full range of pregnancy the relationship curves, so growth charts are built from more flexible models than a single straight line.

Third, cardiovascular risk. Tools like the Framingham Risk Score combine age, blood pressure, and other factors to estimate a patient's 10-year risk of heart disease, and they guide decisions like whether to start a statin. They're built from close relatives of linear regression that are designed for outcomes like "did an event happen, and when." The core idea is the same: weigh predictors to produce a prediction.

And finally, birth weight and adult blood pressure. Research suggests that each extra kilogram of birth weight is associated with a slightly lower adult systolic pressure, about 1 to 1.5 points. That's a real and consistent finding, but birth weight explains well under one percent of the variation in adult blood pressure.

A common mistake is to think that means the finding doesn't matter. A low R-squared doesn't mean there's no effect. It means many other things also influence the outcome. A small shift across an entire population can still matter for public health, even if it's useless for predicting one person.

[Pause prompt.] Pause and sort these four: which would you trust to predict a single patient's value?

[Pause 2 seconds.]

HbA1c to glucose, clearly. The others describe real patterns, but they need more information, or more flexible models, before they're precise enough for an individual.

0:52 · Slide 9 — Python in practice, with the notebook code (5 min)

On screen: start on slide 9 for its three steps, then cut to the notebook in `marimo edit` so the code is visible.

Let's look at the code behind everything we've done. There are three steps, and you've already seen each of them in action.

[Cut to the notebook in edit mode, scrolled to the cell under "4 · Evaluating the fit" that begins `clinic = make_patients(...)`.]

Step one: fit the line. This is the line that does it: `stats.linregress`, with BMI as x and systolic pressure as y. It hands back five values. We use the slope, the intercept, and r. The other two are for another lesson.

Step two: predictions and residuals. Two lines. The predicted pressure is the intercept plus the slope times BMI, computed for every patient at once. The residual is the observed pressure minus the predicted one. That's e equals y minus $\hat{y}$, written in Python.

[Scroll to the residual table cell: `clinic[["bmi", "systolic_bp", "predicted_bp", "residual"]]`.]

Here's the result, the same table that's on the slide. Our first patient has a BMI of 24.1 and a measured pressure of 131. The line predicts 127.1, so the residual is plus 3.9: slightly underpredicted.

[Scroll to the plotting cell under section 4.]

Step three: plot it. Scatter the observed patients, draw the line through the predictions, label the axes. The notebook's version adds the residual plot next to it, and I'd encourage you to always make both.

[Cut back to the slides.]

If you'd rather start from working code, everything you've seen today is in the notebook posted with this video.

0:57 · Slide 10 — Agenda and assignment (2 min)

On screen: slide 10, with "Today's Class" reworded for the video (see Slide fixes before recording).

Let's bring it back to where we started. Our patient with a BMI of 28. Our line predicts a systolic pressure of about 135, typically off by about 9 points either way. Compare that with the guess you wrote down at the beginning. How close were you? More importantly, you now know how far off a prediction like that is likely to be, which a guess alone never tells you.

Today we covered the regression line and how to read it, slope and intercept in clinical units, residuals, the least squares method, and how to judge a fit with R-squared, RMSE, and residual plots.

Before the next session: open the notebook and work through the four "Try it yourself" questions at the bottom. They take about ten minutes and they'll make Problem Set 13 much easier. Then complete Problem Set 13 and Quiz 13 on Canvas, and read Chapter 17 on classification. Next week we move from predicting a number to predicting a category, using nearest neighbors.

0:59 · Slides 11–12 — Conclusion and close (1 min)

On screen: the conclusion slide (you on camera), then "Thank You For Watching".

If you remember one thing from today, make it this: a regression line will always give you an answer. Your job is to check whether the answer deserves to be trusted. Look at the residuals, know your RMSE, and stay inside the range of your data.

[Advance to slide 12.]

Thanks for watching. I'll see you next week.

Slide fixes before recording

The script above assumes these corrections. Slide 8 matters most, because an error in a recording outlasts one in a live class.

Slide	Fix	Why
4	Remove the stray "Mean stay: 4.2 days / Mean stay: 5.8 days" labels	Leftover from a length-of-stay template
4	Drop "holding all else constant"	With one predictor nothing else is held; the phrase belongs to multiple regression

Slide	Fix	Why
7	"off by about 8 mmHg on average" → "typically off by about 8 mmHg"	RMSE is not the plain average miss
8	HbA1c → mean glucose: R^2 0.87 → 0.84	ADAG study: $eAG = 28.7 \times HbA1c - 46.7$, $R^2 = 0.84$ (NGSP)
8	Birth weight → adult SBP: remove $R^2 = 0.18$; show a negative slope and " $R^2 < 0.01$ "	About -1.25 mmHg per kg (Lawlor et al., Am J Epidemiol 2002)
8	Framingham: reword to "regression-based risk models" and remove the R^2	The score comes from survival (Cox) and logistic models, which don't report an R^2
8	Gestational age → birth weight ($R^2 = 0.74$): add a citation	I couldn't confirm the figure; the relationship also curves across the full range
9	Put each Step 3 plotting call on its own line	<code>plt.legend()</code> <code>plt.xlabel(...)</code> on one line is a syntax error if pasted
10	Reword "Today's Class" for a video (e.g. "In-class activity" → "After the video: notebook challenge")	There is no live class

Connection to the required reading

Students arrive having seen regression through **standard units and correlation** with Galton's heights; the deck switches to **raw units, scipy, and health data**. Name that bridge out loud in the first ten minutes: same line, two routes to it.

Reading section	What students saw	Where it lands in the deck
15.1 Correlation	r = average of products of standard units; <code>standard_units</code> , <code>correlation</code> ; hybrid cars	Slide 6 slope formula, slide 7 $R^2 = r^2$
15.2 The Regression Line	In standard units, estimate of $y = r \cdot x$; regression effect; <code>slope</code> , <code>intercept</code> , <code>fit</code> ; family heights	Slides 3–4

Reading section	What students saw	Where it lands in the deck
15.3 Method of Least Squares	RMSE of any line; <code>minimize</code> finds the best slope and intercept; Little Women	Slide 6, demo part 1
15.5 Visual Diagnostics	Residual plots: no pattern, curve (nonlinearity), fan (heteroscedasticity); dugongs	Slides 5 and 7, demo part 2
15.6 Numerical Diagnostics	Mean residual = 0; SD of residuals = $\sqrt{(1 - r^2) \cdot \text{SD}(y)}$; $\text{var}(\text{fitted})/\text{var}(y) = r^2$	Slides 5 and 7

The two formulas worth putting side by side on the board:

$$\text{slope} = r \cdot \frac{\text{SD}_y}{\text{SD}_x}, \quad \text{intercept} = \bar{y} - \text{slope} \cdot \bar{x}$$

$$\text{SD}(\text{residuals}) = \sqrt{1 - r^2} \cdot \text{SD}(y)$$

The second one is the book's version of RMSE on slide 7, so students can predict the RMSE from r before the notebook computes it. Two vocabulary notes: the book uses `fit` and "fitted values" where the deck uses $\hat{y}$ and `predicted_bp`, and the book's `minimize` does by search what `scipy.stats.linregress` does in closed form.

Screen-recorded marimo demo

The notebook `linear_regression_demo.py` runs top to bottom without errors and follows the deck's order. With the default settings it fits $\hat{y} = 76.3 + 2.11 \cdot \text{BMI}$ with $R^2 = 0.53$ and RMSE = 9.2 mmHg, close to slide 7's example numbers.

Setup (do once before class)

```
pip install marimo numpy pandas scipy matplotlib
marimo edit linear_regression_demo.py    # editable, code visible
marimo run linear_regression_demo.py     # app view, code hidden
```

Screen-record demo parts 1 and 2 in `marimo run` (code hidden, bigger plots), and slide 9 in `marimo edit` so the code shows. Zoom the browser to 125–150% so sliders and numbers are readable in the video. Post the notebook with the video so viewers can follow along and do the challenge afterward.

The data: 200 simulated patients generated from slide 4's model ($68.5 + 2.4 \cdot \text{BMI}$) plus noise. Say "simulated" out loud. The column names (`bmi` , `systolic_bp` , `predicted_bp` , `residual`) match slide 9 exactly.

Answers to the try-it-yourself questions (post these after the due date, or narrate one as a worked example)

1. Noise 4 $\rightarrow$ 20: R^2 falls from about 0.89 to 0.17 and RMSE rises from about 4 to 18 mmHg, but the slope stays in the same neighborhood ($2.3 \rightarrow 1.8$ on the default sample). Noise changes how well we predict, not what the trend is.
2. Curved: R^2 is about 0.40, still respectable-looking, yet the residual plot shows a clear U. The line is the wrong shape; R^2 alone doesn't reveal that.
3. The extreme patient pulls the slope from about 2.11 to 1.85 (12%). Check for a data entry error or measurement problem first; delete only with a documented reason, and report results both ways.
4. A low R^2 doesn't mean no effect (the slope can be real and clinically meaningful), and regression on observational data can't show that BMI *affects* BP at all.

Recording tips: reset the sliders to their defaults before each take (a page refresh does it), and keep the noise at 10 and the sample unchanged so the numbers you say match the numbers in this doc. If a take goes wrong, refresh and redo just that section.

Common misconceptions

Six misconceptions come up reliably with this material. With no one in the room to voice them, raise each one yourself ("A common mistake here is...") and correct it on camera; each has a notebook moment that shows why it's wrong.

Misconception	Correction	Where to show it
"The slope means raising a patient's BMI raises their BP."	Slope compares patients; with observational data it is association, not causation (15.1).	Slide 4; try-it-yourself Q4
"Low R^2 means no relationship."	A real slope can explain little variance; birth weight and adult BP is the textbook case.	Slide 8

Misconception	Correction	Where to show it
"High R^2 means the line is right."	A curved relationship can still score a decent R^2 ; only the residual plot shows the misfit.	Notebook section 4, Curved
"Residuals are the shortest distance to the line."	They are vertical: we are predicting y , so error is measured in y .	Notebook section 1
"The intercept is the BP of a patient with BMI 0."	It is an anchor far outside the data; predictions there are extrapolation.	Notebook section 5, BMI = 0
"Outliers should just be deleted."	Investigate first; one point can move the slope 12% in the demo, so report how much it matters.	Notebook section 4, extreme patient

Wrap-up and check questions

Check questions (pose one or two on camera as pause prompts and give the answer, or move them into Quiz 13 on Canvas)

1. A model predicts HbA1c 7% $\rightarrow$ eAG 154 mg/dL. A patient's measured mean glucose is 170. What is the residual, and is the model over- or underpredicting? (+16 mg/dL; underpredicting.)
2. In one sentence, why does least squares square the residuals instead of just adding them? (So positive and negative errors can't cancel out, and large misses count more.)
3. $r = 0.5$ between two variables. What fraction of the variance in y does the line explain? (25%.)

Before next class (from slide 10): read Chapter 17 (Classification), complete Problem Set 13 and Quiz 13 on Canvas.

Pre-recording checklist

- ☐ Remove the stray "Mean stay" labels on slide 4 and the phrase "holding all else constant"
- ☐ Correct or cite the four R^2 values on slide 8 and reword the Framingham example
- ☐ Split the slide 9 Step 3 plotting line into separate lines

- ☐ Reword slide 10's live-class items ("In-class activity") for a video
- ☐ Install marimo on the recording machine, zoom the browser, and do a test take of demo part 1
- ☐ Post `linear_regression_demo.py` alongside the video for the try-it-yourself questions

Sources

- [Inferential Thinking, Ch. 15 — Prediction](#) and sections [15.1](#), [15.2](#), [15.3](#), [15.5](#), [15.6](#)
- [NGSP: HbA1c and estimated average glucose \(ADAG formula, \$R^2 = 0.84\$ \)](#)
- [Lawlor, Ebrahim, Davey Smith. Birth weight and later systolic BP meta-regression, Am J Epidemiol 2002](#)
- Slide deck: M10_01_CAI5735_su26_Aggarwal_v1 (provided)